Thick Reality Front: a numerical framework for spacetime structure in quantum evolution

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We introduce a computational framework for exploring a "thick reality front" (TRF) description of quantum dynamics. Instead of associating physical structure with a single instantaneous wavefunction, the framework assumes that observable patterns may emerge from a finite temporal region surrounding the present.

Forward quantum evolution from preparation is combined with backward propagation conditioned on detection events. Their local amplitude overlap produces a spacetime field that highlights regions simultaneously compatible with preparation and measurement.

Numerical simulations of a double-slit experiment demonstrate that this overlap field often develops coherent ridge structures that trace narrow probability flow channels through spacetime.

Baseline simulations without the thick-front operator are also presented for comparison, allowing the influence of the TRF framework on branch structure and ridge stability to be examined.

The framework reproduces standard quantum predictions while providing a geometric visualization of probability flow between preparation and measurement.

INTRODUCTION

Quantum mechanics describes the evolution of a system through the Schrödinger equation

$$i\hbar\partial_t\psi = H\psi. \tag{1}$$

While the equation determines the unitary evolution of the wavefunction, measurement outcomes appear to select definite results from a superposition of possibilities.

Several approaches attempt to address this issue, including collapse models, Bohmian mechanics, and time-symmetric formulations of quantum theory. Time-symmetric approaches suggest that both preparation and measurement conditions may play roles in shaping the structure of quantum processes.

In this work we explore a computational framework inspired by this idea. Rather than associating physical structure with a single time slice of the wavefunction, we assume that observable structure may emerge from a finite temporal region surrounding the present. We refer to this region as a *thick reality front*.

The purpose of this work is not to propose a new fundamental dynamical law but to develop a numerical framework for analyzing forward–backward amplitude structures and the spacetime patterns that emerge from them.

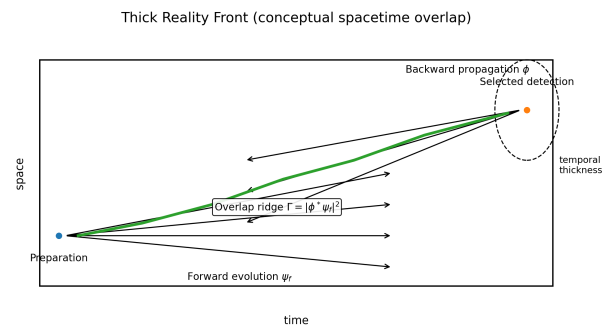

FIG. 1. Conceptual illustration of the thick reality front (TRF). Forward evolution from the preparation event produces a spreading wavefunction ψ_f. Backward propagation from a selected detection event produces a retrodictive wave ϕ. Their local amplitude overlap $\Gamma = |\phi^*\psi_f|^2$ highlights spacetime regions supported simultaneously by both conditions.

RELATED WORK

Several recent approaches explore how measurement outcomes or effective structure can emerge from quantum systems, but they typically locate the selection mechanism either in the measurement process itself or in engineered system dynamics.

Measurement-based approaches treat detectors or measurement architectures as feature maps that transform quantum states into classical data representations [2]. In such models, outcome selection is effectively implemented at the readout stage through a stable measurement mapping that can, in some cases, be calibrated using classical inputs. While this provides an efficient framework for quantum state inference, it does not address where or how the selection of outcomes originates within the underlying quantum dynamics.

A different class of models introduces explicit attractor structures through engineered system design. For example, quantum generalizations of Hopfield neural networks implement associative memory through dissipative and coherent dynamics [3], where stored patterns correspond to attractors of the system. These models demonstrate that open quantum dynamics can support nontrivial steady states and even limit-cycle behavior arising

from the interplay of coherent evolution and dissipation. However, in such constructions the attractor landscape is explicitly encoded in the Hamiltonian and jump operators, and selection corresponds to convergence toward predefined states.

The present work differs from both perspectives in that it does not place selection at the measurement stage nor encode it in a predefined attractor landscape. Instead, the framework explores whether structured, localized propagation channels can emerge directly from the overlap of forward and backward amplitudes in spacetime. In this view, selection is associated with dynamical structure formation during propagation rather than with readout or engineered convergence.

This distinction motivates the central question addressed here: whether outcome-relevant structures can arise as emergent features of quantum evolution itself, without being reducible to detector response functions or predesigned attractor dynamics.

BASELINE QUANTUM EVOLUTION

Before introducing the TRF framework it is useful to examine the baseline quantum dynamics.

Forward evolution from an initial state ψ_0 is given by

$$\psi_f(t) = U(t, t_0)\psi_0. \tag{2}$$

The corresponding probability density is

$$\rho_f(x, t) = |\psi_f(x, t)|^2. \tag{3}$$

In numerical simulations of a double-slit geometry this forward evolution produces the familiar interference pattern on a downstream detection screen.

The baseline dynamics may already exhibit structured probability flow patterns upstream of the detection screen. Understanding these structures provides an important reference for evaluating the additional effects introduced by the thick-front framework.

FORWARD AND BACKWARD EVOLUTION

Suppose a detection occurs at position x_d and time t_d. We construct a localized detection state

$$\phi(x, t_d) = \exp\left(-\frac{(x - x_d)^2}{2\sigma_c^2}\right). \tag{4}$$

Backward propagation is defined through the adjoint evolution

$$\phi(t) = U^\dagger(t_d, t)\phi(t_d). \tag{5}$$

The forward and backward states can therefore be defined at every intermediate spacetime point.

AMPLITUDE OVERLAP FIELD

The central quantity in the TRF framework is the local amplitude overlap

$$A(x, t) = \phi^*(x, t)\psi_f(x, t). \tag{6}$$

The ridge intensity field is then defined as

$$\Gamma(x, t) = |A(x, t)|^2. \tag{7}$$

Large values of Γ identify spacetime regions that are simultaneously compatible with both preparation and measurement conditions.

In numerical experiments this field frequently develops narrow high-intensity structures forming continuous paths through spacetime.

TEMPORAL THICKNESS

Instead of using a single detection time we allow a temporal distribution of possible detection times within a window of width σ_T.

Backward states are therefore averaged as

$$E_{\text{mix}}(x, t) = \sum_k w_k |\phi_k(x, t)|^2, \tag{8}$$

with Gaussian weights

$$w_k = \frac{\exp[-(t_k - t_d)^2/(2\sigma_T^2)]}{\sum_j \exp[-(t_j - t_d)^2/(2\sigma_T^2)]}. \tag{9}$$

This produces a temporally extended retrodictive weighting representing the finite thickness of the reality front.

RIDGE EXTRACTION

The overlap field $\Gamma(x, t)$ often develops narrow high-intensity structures.

The simplest ridge trajectory is defined by

$$\mathbf{r}_R(t) = \arg\max_{\mathbf{r}} \Gamma(\mathbf{r}, t). \tag{10}$$

More stable ridge definitions include centroid estimation within the top intensity fraction, snapping to nearby local maxima, and local maximum tracking across frames.

These methods improve temporal continuity when multiple competing maxima are present.

VELOCITY FIELD COMPARISON

The quantum probability current is

$$\mathbf{j} = \frac{\hbar}{m}\mathrm{Im}(\psi^* \nabla \psi). \tag{11}$$

This defines the velocity field

$$\mathbf{v} = \frac{\mathbf{j}}{\rho_f}. \tag{12}$$

Bohmian trajectories constructed from this velocity field provide a useful comparison with ridge trajectories extracted from the overlap field.

THICK-FRONT OPERATOR

The numerical framework can optionally include a *thick-front operator* that modifies the forward evolution according to local coherence structure.

The purpose of this operator is not to replace the Schrödinger dynamics but to explore how a finite spacetime thickness of physical structure might influence branch competition and ridge formation.

The specific numerical form used in the present simulations includes

- local coherence sharpening
- optional branch competition between nearby maxima
- normalization preserving overall probability

The precise functional form used in the present experiments is given in Ref. [placeholder] or the accompanying code repository [1].

NUMERICAL EXPERIMENT

We simulate a two-dimensional double-slit experiment using the split-operator method.

A Gaussian wave packet propagates toward a barrier containing two narrow slits and forms an interference pattern on a downstream screen.

Forward evolution generates the interference pattern while backward propagation from a selected click state produces a retrodictive field.

The overlap field Γ then highlights spacetime channels connecting the preparation region with the selected detection event.

BASELINE VERSUS TRF COMPARISON

To evaluate the influence of the thick-front operator we perform simulations both with and without this modification.

Baseline simulations reproduce standard Schrödinger dynamics. The TRF simulations introduce the additional coherence-based operator described above.

Qualitative comparisons focus on

- ridge sharpness
- stability of ridge trajectories
- suppression or enhancement of nearby competing branches
- agreement with Bohmian trajectory structures

Representative results are shown in Fig. [placeholder].

In preliminary experiments the TRF framework tends to emphasize narrow ridge-like probability channels while suppressing some nearby competing structures.

LIMITATIONS

The present framework is exploratory and several limitations should be noted.

- numerical results depend on grid resolution
- competition operators may introduce discretization artifacts
- the framework does not yet include a full detector model
- the thick-front operator is currently phenomenological

Future work will investigate more physically motivated implementations and improved numerical stability.

CONCLUSION

We introduced a computational framework for exploring thick reality front descriptions of quantum dynamics.

By combining forward evolution with backward propagation from detection events the method produces structured spacetime overlap fields.

These fields frequently reveal coherent ridge structures that trace probability flow between preparation and measurement.

Baseline simulations demonstrate that similar structures may already exist in the underlying quantum dynamics, while the TRF framework modifies their prominence and stability.

The framework provides a geometric visualization of spacetime probability structure and may offer a useful tool for exploring time-symmetric aspects of quantum evolution.

[1] Saku Hämäläinen. Quantumtoy: Thick reality front simulation framework, 2026. Code repository (to be made public).

[2] Jan Ludwig, Matteo Marcucci, Igor Lesanovsky, et al. Quantum inference on a classically trained quantum extreme learning machine. 2026.

[3] P. Rotondo, M. Marcuzzi, J. P. Garrahan, and I. Lesanovsky. Open quantum generalisation of hopfield neural networks. *Journal of Physics A: Mathematical and Theoretical*, 50(37):375301, 2017.